

swigh-score (hyperthreaded) quickstart

Setup

- Ensure `bin/SWscore_hyperthreaded` is built (`gcc -O2 -pthread -o bin/SWscore_hyperthreaded bin/SWscore_hyperthreaded.c`).
- FASTQ test file is available (e.g., `gzip -dc test.fastq.gz > test.fastq`).

Core commands

- Align clonotype to reads:
 - `./swigh-score -i <file.fastq> -r <CLONOTYPE> -t <threads> [-o prefix] [-B batch] [-b band] [--no-dedup]`
 - Positional legacy still works: `./swigh-score <file.fastq> <threads> <CLONOTYPE> [prefix]`
 - Outputs: `tmp/<file>.SWscore.out` (raw scores), `<prefix><file>.out` (joined with counts).
- Report similarity distribution and top V/J from the `.out` file:
 - `./swigh-report <file.fastq.out>`
- Random clonotyping of 1000 reads (default V-gene):
 - `./swigh-clonotype-random <file.fastq> <threads> [vgene|jgene]`
 - Outputs `<file>.<gene>.txt` with counts.
- Amplicon clonotyping (most recurrent sequence):
 - `./swigh-clonotype-amplicon <file.fastq> [optional_read_count]`
- Gene matches for a clonotype sequence:
 - `./swigh-clonotype-sequence <CLONOTYPE> [top_n]`

Notes

- Thread count defaults to detected cores if `-t` is omitted.
- `-b/--band` sets banded Smith–Waterman half-width; 0 = full SW.
- Output columns from the scorer match legacy format: `<read> <similarity> <SW score> <aligned bases> <swigh-score>`.
- Pandoc helper: if you want a PDF/HTML version, activate the bundled env and run, e.g., `source pandoc_env/bin/activate && pandoc -s HYPERTHREADED_USAGE.md -o HYPERTHREADED_USAGE.pdf`.
- `--no-dedup` skips collapsing duplicate reads (useful on very diverse datasets); default is to deduplicate for speed on clonal data.